Correcting errors in the side-channel attacks – algorithm visualization

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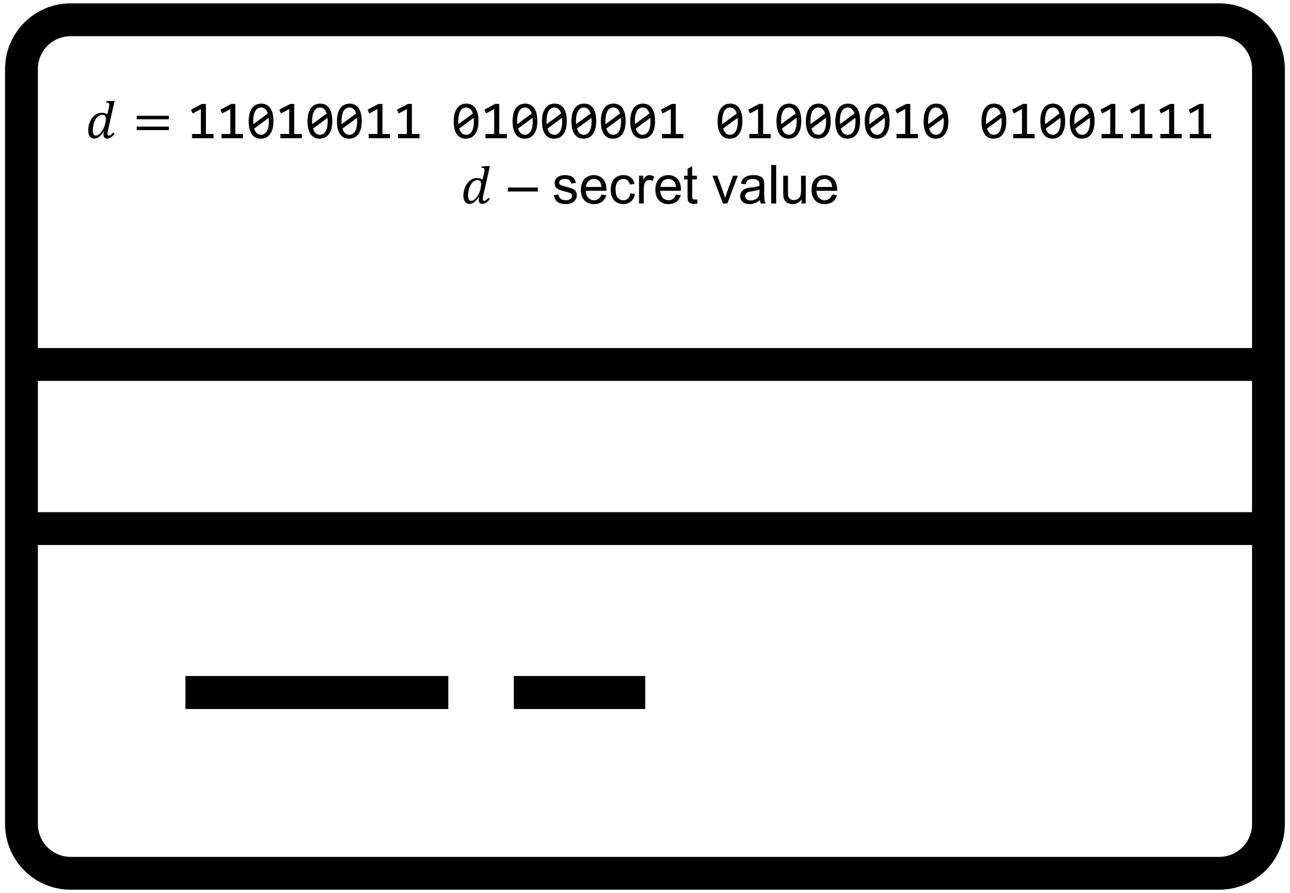
Smart card



Secret scalar

$d =$ `11010011 01000001 01000010 01001111`

Legend
d – secret value

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Secret scalar blinding

$d =$ `11010011 01000001 01000010 01001111`

$E =$ `00000001 00000000 00000000 01000001 01001011`

Legend
d – secret value
E – group order

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Secret scalar blinding

$d =$	`11010011 01000001 01000010 01001111`
$E =$	`00000001 00000000 00000000 01000001 01001011`
$r_l =$	`11001010`

Legend
d – secret value
E – group order
r_l – blinding factor

Secret scalar blinding

$$
\begin{array}{rr}
d = & 11010011\ 01000001\ 01000010\ 01001111 \\
E = & 00000001\ 00000000\ 00000000\ 01000001\ 01001011 \\
r_l = & 11001010 \\
\hline
+ E \times r_l = & 11001010\ 00000000\ 00110011\ 10000101\ 00101110
\end{array}
$$

Legend

d – secret value

E – group order

r_l – blinding factor

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Secret scalar blinding

$d =$	11010011 01000001 01000010 01001111
$E =$	00000001 00000000 00000000 01000001 01001011
$r_l =$	11001010
$+ E \times r_l =$	11001010 00000000 00110011 10000101 00101110
$d + E \times r_l =$	11001010 11010011 01110100 11000111 01111101

Legend

d – secret value

E – group order

r_l – blinding factor

Secret scalar blinding

$$
\begin{array}{rr}
d = & 11010011\ 01000001\ 01000010\ 01001111 \\
E = & 00000001\ 00000000\ 00000000\ 01000001\ 01001011 \\
r_l = & \color{red}{11001010} \\
\hline
+E \times r_l = & \color{red}{11001010}\ 00000000\ 00110011\ 10000101\ 00101110 \\
\hline
d + E \times r_l = & \color{red}{11001010}\ 11010011\ 01110100\ 11000111\ 01111101
\end{array}
$$

Legend
d – secret value
E – group order
r_l – blinding factor

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Secret scalar blinding

$d =$		11010011	01000001	01000010	01001111
$E =$	00000001	00000000	00000000	01000001	01001011
$r_l =$					11001010
$+ E \times r_l =$	11001010	00000000	00110011	10000101	00101110
$d + E \times r_l =$	11001010	11010011	01110100	11000111	01111101

Legend

d – secret value

E – group order

r_l – blinding factor

Secret scalar blinding

$d =$		11010011	01000001	01000010	01001111
$E =$	00000001	00000000	00000000	01000001	01001011
$r_l =$					11001010
$+ E \times r_l =$	11001010	00000000	00110011	10000101	00101110
$d + E \times r_l =$	11001010	11010011	01110100	11000111	01111101

Legend
d – secret value
E – group order
r_l – blinding factor

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Secret scalar blinding

$d =$		11010011	01000001	01000010	01001111
$E =$	00000001	00000000	00000000	01000001	01001011
$r_l =$					11001010
$+ E \times r_l =$	11001010	00000000	00110011	10000101	00101110
$d + E \times r_l =$	11001010	11010011	01110100	11000111	01111101
$E_0 =$	00000000	00000000	00000000	01000001	01001011

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Secret scalar blinding

$d =$		11010011	01000001	01000010	01001111
$E =$	00000001	00000000	00000000	01000001	01001011
$r_l =$					11001010
$+ E \times r_l =$	11001010	00000000	00110011	10000101	00101110
$d + E \times r_l =$	11001010	11010011	01110100	11000111	01111101
$E_0 =$	00000000	00000000	00000000	01000001	01001011
$d_M = 2^K \times r =$	11001010	00000000	00000000	00000000	00000000
$d_L = E_0 \times r =$	00000000	11010011	01110100	11000111	01111101

Secret scalar blinding – traces

$d =$ 11010011 01000001 01000010 01001111

$r_1 =$ 11001010

$r_2 =$ 10010001

Secret scalar blinding – traces

$$
\begin{array}{rr}
d = & \texttt{11010011 01000001 01000010 01001111} \\
r_1 = & \texttt{11001010} \\
r_2 = & \texttt{10010001} \\
\hline
d_1 = & \texttt{11001010 11010011 01110100 11000111 01111101} \\
d_2 = & \texttt{10010001 11010011 01100110 00111101 11001010}
\end{array}
$$

Secret scalar blinding – traces

$d =$		11010011	01000001	01000010	01001111
$r_1 =$					11001010
$r_2 =$					10010001
$d_1 =$	11001010	11010011	01110100	11000111	01111101
$d_2 =$	10010001	11010011	01100110	00111101	11001010

Secret scalar blinding – traces

$d =$	11010011 01000001 01000010 01001111
$r_1 =$	11001010
$r_2 =$	10010001
$d_1 =$	11001010 11010011 01110100 11000111 01111101
$d_2 =$	10010001 11010011 01100110 00111101 11001010
$\widetilde{d_1} = d_1 \oplus \varepsilon_1 =$	11001010 11010011 01110100 11000111 01111101
$\widetilde{d_2} = d_2 \oplus \varepsilon_2 =$	10010001 11010011 01100110 00111101 11001010

Secret scalar blinding – traces

$d =$	11010011 01000001 01000010 01001111
$r_1 =$	11001010
$r_2 =$	10010001
$d_1 =$	11001010 11010011 01110100 11000111 01111101
$d_2 =$	10010001 11010011 01100110 00111101 11001010
$\widetilde{d_1} = d_1 \oplus \varepsilon_1 =$	11001110 10010011 01010100 11001111 00101101
$\widetilde{d_2} = d_2 \oplus \varepsilon_2 =$	10000001 11000111 01100010 01111001 11001010

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Secret scalar blinding – traces

$$d = \texttt{11010011 01000001 01000010 01001111}$$

$$\widetilde{d_1} = \texttt{11001110 10010011 01010100 11001111 00101101}$$

$$\widetilde{d_2} = \texttt{10000001 11000111 01100010 01111001 11001010}$$

Algorithm

$$
\begin{array}{rl}
d = & \texttt{ 11010011 01000001 01000010 01001111} \\
\widetilde{d_1} = & \texttt{11001110 10010011 01010100 11001111 00101101} \\
\widetilde{d_2} = & \texttt{10000001 11000111 01100010 01111001 11001010} \\
\hline
\end{array}
$$

Algorithm – preparation

$$d = \quad \texttt{11010011 01000001 01000010 01001111}$$
$$\widetilde{d_1} = \texttt{11001110 10010011 01010100 11001111 00101101}$$
$$\widetilde{d_2} = \texttt{10000001 11000111 01100010 01111001 11001010}$$

window $t = 3$, preserved canidates $L = 2$

Lr_i		
$\tilde{r}_1$	00000000	-1
$\tilde{r}_2$	00000000	-1

$$d^* = 0$$

Algorithm – step 1

$$d = \quad\quad\quad\quad \texttt{11010011 01000001 01000010 01001111}$$
$$\widetilde{d_1} = \texttt{11001110 10010011 01010100 11001111 00101101}$$
$$\widetilde{d_2} = \texttt{10000001 11000111 01100010 01111001 11001010}$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

Lr_i		
$\tilde{r}_1$	00000000	-1
$\tilde{r}_2$	00000000	-1

$$d^* = 0$$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0	0

$\bar{\mathrm{d}}$	
00000000	00000001

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Algorithm – step 1

$$d = \qquad\qquad 11010011\ 01000001\ 01000010\ 01001111$$

$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$

$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

Lr_i		
$\tilde{r}_1$	00000000	-1
$\tilde{r}_2$	00000000	-1

$Lr_{i-\mathrm{temp}}$				
$\bar{r}_1$	00000000	00000001		
$\bar{r}_2$	00000000	00000001		

Algorithm – step 1

$$
\begin{aligned}
d &= 11010011\ 01000001\ 01000010\ 01001111 \\
\widetilde{d_1} &= 11001110\ 10010011\ 01010100\ 11001111\ 00101101 \\
\widetilde{d_2} &= 10000001\ 11000111\ 01100010\ 01111001\ 11001010
\end{aligned}
$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0	0

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\mathrm{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

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Algorithm – step 1

$$d = \qquad\qquad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0	0

$\bar{\mathrm{d}}$
00000000

$Lr_{i-\mathrm{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

$$\bar{d}_l = \bar{\mathrm{r}}_1 \times E + \bar{\mathrm{d}}$$

Algorithm – step 1

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0	0

$\bar{\mathrm{d}}$	
00000000	

Lr_{i-temp}				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

$$\bar{d_l} = \bar{\mathrm{r}}_1 \times E + \bar{\mathrm{d}}$$

```
                                              00000000
× 00000001 00000000 00000000 01000001 01001011
+                                             00000000
                                              00000000
```

Algorithm – step 1

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$

$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$

$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{d} = 0$	$\hat{d} = 1$
0	0

$\bar{d}$
00000000

Lr_{i-temp}				
$\bar{r}_1$	00000000	00000001		
$\bar{r}_2$	00000000	00000001		

$$\bar{d}_l = 00000000$$

Algorithm – step 1

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $t = 3$, preserved canidates $L = 2$

$\hat{d} = 0$	$\hat{d} = 1$
0	0

$\bar{d}$	
00000000	00000001

$Lr_{i-\text{temp}}$				
$\bar{r}_1$	00000000	00000001		
$\bar{r}_2$	00000000	00000001		

$$\bar{d}_l = 0000000[0]$$

i

Algorithm – step 1

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$

$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$

$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{d} = 0$	$\hat{d} = 1$
0	0

$\bar{d}$	
00000000	00000001

$Lr_{i-\text{temp}}$				
$\bar{r}_1$	00000000	00000001		
$\bar{r}_2$	00000000	00000001		

$$d_l^t = 00000[000]$$

t

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Algorithm – step 1

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0	0

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\text{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

$$d_l^t = \quad 000$$

Algorithm – step 1

$$d = 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0	0

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\mathrm{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

$$d_l^t = 000$$
$$\bar{\mathrm{r}}_1 = 00000000$$

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Algorithm – step 1

$$d = \quad \texttt{11010011 01000001 01000010 01001111}$$

$$\widetilde{d_1} = \texttt{11001110 10010011 01010100 11001111 00101101}$$

$$\widetilde{d_2} = \texttt{10000001 11000111 01100010 01111001 11001010}$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0	0

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\mathrm{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

$$d_l^t = \texttt{000}$$

$$\mathrm{r}_l^t = \texttt{00000}\underbrace{\texttt{[000]}}_{\mathrm{t}}$$

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Algorithm – step 1

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0	0

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\mathrm{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

$$d_l^t = 000$$
$$\mathrm{r}_l^t = 000$$

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Algorithm – step 1

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$

$$\widetilde{d_1} = 1100111[0]\ 10010011\ 01010100\ 11001111\ 0010110[1]$$

$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0	0

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\mathrm{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

$$d_l^t = 00[0]$$

$$\mathrm{r}_l^t = 00[0]$$

Algorithm – step 1

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$

$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$

$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0	0

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\mathrm{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

$$d_l^t = 0$$

$$\mathrm{r}_l^t = 0$$

Algorithm – step 1

$d =$ 11010011 01000001 01000010 01001111

$\widetilde{d_1} =$ 11001110 10010011 01010100 11001111 00101101

$\widetilde{d_2} =$ 10000001 1100[illegible]111 01100010 01111001 110010[illegible]0

itera[illegible] $i = 1$, window $\mathrm{t} = 3$, preserved canidates [illegible] $= 2$

HD()

HD()

$d_l^t =$ 0

$\mathrm{r}_l^t =$ 0

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0	0

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\mathrm{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

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Algorithm – step 1

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$

$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$

$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0	0

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\text{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

$$HD(0,0) = 0$$

$$HD(0,1) = 1$$

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Algorithm – step 1

$$d = \qquad\qquad 11010011\ 01000001\ 01000010\ 01001111$$

$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$

$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0	0

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\mathrm{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

$$\mathrm{h} = \sum \begin{matrix} HD(0,0) = 0 \\ HD(0,1) = 1 \end{matrix}$$

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Algorithm – step 1

$$d = \quad\quad\quad\quad\quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0	0

$\bar{\mathrm{d}}$	
00000000	00000001

$$h = 1$$

$Lr_{i-\text{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

Algorithm – step 1

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{d} = 0$	$\hat{d} = 1$
0	0

$\bar{d}$	
00000000	00000001

$Lr_{i-\mathrm{temp}}$				
$\bar{r}_1$	00000000	00000001		
$\bar{r}_2$	00000000	00000001		

$$h = 1$$
$$\mathrm{p} = \epsilon_b^h \times (1 - \epsilon_b)^{2w-h}$$

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Algorithm – step 1

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0	0

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\mathrm{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

$$h = 1$$
$$\mathrm{p} = \epsilon_b^h \times (1 - \epsilon_b)^{2w-h}$$
$$\mathrm{p} = 0.15^1 \times 0.85^{2-1}$$
$$\mathrm{p} = 0.1275$$

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Algorithm – step 1

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0	0

$\bar{\mathrm{d}}$	
00000000	00000001

$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

+

$$h = 1$$
$$\mathrm{p} = \epsilon_b^h \times (1 - \epsilon_b)^{2w-h}$$
$$\mathrm{p} = 0.15^1 \times 0.85^{2-1}$$
$$\mathrm{p} = 0.1275$$

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Algorithm – step 1

$$d = \quad\quad\quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0.1275	0

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\mathrm{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

Algorithm – step 1

$$d = \qquad\qquad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0.1275	0

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\mathrm{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

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Algorithm – step 1

$$d = \quad\quad\quad\quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0.255	0

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\mathrm{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

Algorithm – step 1

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0.3825	0

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\mathrm{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

Algorithm – step 1

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0.51	0

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\text{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

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Algorithm – step 1

$$d = \quad\quad\quad\quad\quad 11010011\ 01000001\ 01000010\ 01001111$$

$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$

$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0.51	0.7225

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\mathrm{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

Algorithm – step 1

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0.51	0.745

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\mathrm{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

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Algorithm – step 1

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$

$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$

$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0.51	0.7675

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\mathrm{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

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Algorithm – step 1

$$d = \quad \texttt{11010011 01000001 01000010 01001111}$$
$$\widetilde{d_1} = \texttt{11001110 10010011 01010100 11001111 00101101}$$
$$\widetilde{d_2} = \texttt{10000001 11000111 01100010 01111001 11001010}$$

---

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0.51	1.49

$\bar{\mathrm{d}}$	
00000000	00000001

$Lr_{i-\text{temp}}$				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

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Algorithm – step 1

$$d = \quad \texttt{11010011 01000001 01000010 01001111}$$

$$\widetilde{d_1} = \texttt{11001110 10010011 01010100 11001111 00101101}$$

$$\widetilde{d_2} = \texttt{10000001 11000111 01100010 01111001 11001010}$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$\hat{\mathrm{d}} = 0$	$\hat{\mathrm{d}} = 1$
0.51	1.49

$0.51 < 1.49$

$d^* = \texttt{00000001}$

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Algorithm – post step 1

$$d = \qquad\qquad\qquad 11010011\ 01000001\ 01000010\ 01001111$$

$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$

$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

Lr_i		
$\tilde{r}_1$	00000000	-1
$\tilde{r}_2$	00000000	-1

$$d^* = 00000001$$

MUNI

Algorithm – step 2

$$d = \texttt{11010011 01000001 01000010 01001111}$$
$$\widetilde{d_1} = \texttt{11001110 10010011 01010100 11001111 00101101}$$
$$\widetilde{d_2} = \texttt{10000001 11000111 01100010 01111001 11001010}$$

iteration $i = 1$, window t $= 3$, preserved canidates L $= 2$

$d^* = \texttt{00000001}$

Lr_i		
$\tilde{r}_1$	00000000	-1
$\tilde{r}_2$	00000000	-1

Lr_{i+1}				
$\bar{r}_1$	00000000	00000001		
$\bar{r}_2$	00000000	00000001		

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Algorithm – step 2

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$d^* = 00000001$

Lr_{i+1}				
$\bar{r_1}$	00000000	00000001		
$\bar{r_2}$	00000000	00000001		

Algorithm – step 2

$$d = \quad\quad\quad\quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

---

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$$d^* = 00000001$$

Lr_{i+1}				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

$$\bar{d}_l = \bar{\mathrm{r}}_1 \times E + d^*$$

Algorithm – step 2

$$
\begin{aligned}
d &= 11010011\ 01000001\ 01000010\ 01001111 \\
\widetilde{d_1} &= 11001110\ 10010011\ 01010100\ 11001111\ 00101101 \\
\widetilde{d_2} &= 10000001\ 11000111\ 01100010\ 01111001\ 11001010
\end{aligned}
$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$d^* = 00000001$

Lr_{i+1}				
$\overline{\mathrm{r}_1}$	00000000	00000001		
$\overline{\mathrm{r}_2}$	00000000	00000001		

$$\overline{d_l} = \overline{\mathrm{r}_1} \times E + d^*$$

```
                                            00000000
× 00000001 00000000 00000000 01000001 01001011
+                                           00000001
                                            00000001
```


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Algorithm – step 2

$$d = \quad \texttt{11010011 01000001 01000010 01001111}$$
$$\widetilde{d_1} = \texttt{11001110 10010011 01010100 11001111 00101101}$$
$$\widetilde{d_2} = \texttt{10000001 11000111 01100010 01111001 11001010}$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$d^* = \texttt{00000001}$

$\bar{d_l} = \texttt{00000001}$

Lr_{i+1}				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

MUNI

Algorithm – step 2

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$

$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$

$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$d^* = 00000001$

Lr_{i+1}				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

$\bar{d}_l =$ 0000000[1]

i

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Algorithm – step 2

$$d = \quad\quad\quad\quad \texttt{11010011 01000001 01000010 01001111}$$
$$\widetilde{d_1} = \texttt{11001110 10010011 01010100 11001111 00101101}$$
$$\widetilde{d_2} = \texttt{10000001 11000111 01100010 01111001 11001010}$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$d^* = \texttt{00000001}$

Lr_{i+1}				
$\bar{\mathrm{r}}_1$	00000000	00000001		
$\bar{\mathrm{r}}_2$	00000000	00000001		

$\bar{d}_l = \texttt{00000001}$

Algorithm – step 2

$$d = \quad \texttt{11010011 01000001 01000010 01001111}$$

$$\widetilde{d_1} = \texttt{1100111[0] 10010011 01010100 11001111 0010110[1]}$$

$$\widetilde{d_2} = \texttt{10000001 11000111 01100010 01111001 11001010}$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$d^* = 00000001$

Lr_{i+1}				
$\bar{r_1}$	0000000[0]	00000001		
$\bar{r_2}$	00000000	00000001		

$\bar{d_l} = \quad 1$

Algorithm – step 2

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$

$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$

$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$$d^* = 00000001$$

Lr_{i+1}				
$\bar{r_1}$	0	00000001		
$\bar{r_2}$	00000000	00000001		

$$\bar{d_l} = \quad 1$$

Algorithm – step 2

$d =$ 11010011 01000001 01000010 01001111

$\widetilde{d_1} =$ 11001110 10010011 01010100 11001111 00101101

$\widetilde{d_2} =$ 1000[illegible]001 11000111 01100010 01111001 110010[illegible]0

---

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidate [illegible] $= 2$

$d^* =$ 00000001

HD()

HD()

$\bar{d_l} =$ 1

	Lr_{i+1}			
$\bar{r_1}$	0	00000001		
$\bar{r_2}$	00000000	00000001		

MUNI

Algorithm – step 2

$$d = \quad\quad\quad\quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$d^* = 00000001$

Lr_{i+1}				
$\bar{r}_1$	00000000	00000001		
$\bar{r}_2$	00000000	00000001		

$HD(0,0) = 0$

$HD(0,0) = 0$

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Algorithm – step 2

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$d^* = 00000001$

Lr_{i+1}				
$\bar{r_1}$	00000000	00000001		
$\bar{r_2}$	00000000	00000001		

$$h = \sum \begin{matrix} HD(0,0) = 0 \\ HD(0,0) = 0 \end{matrix}$$

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Algorithm – step 2

$$d = \quad\quad\quad\quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

---

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$d^* = 00000001$

Lr_{i+1}				
$\bar{r_1}$	00000000	00000001		
$\bar{r_2}$	00000000	00000001		

$h = 0$

Algorithm – step 2

$$d = \quad\quad\quad\quad \texttt{11010011 01000001 01000010 01001111}$$
$$\widetilde{d_1} = \texttt{11001110 10010011 01010100 11001111 00101101}$$
$$\widetilde{d_2} = \texttt{10000001 11000111 01100010 01111001 11001010}$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$$d^* = \texttt{00000001}$$

Lr_{i+1}				
$\bar{r}_1$	00000000	00000001		
$\bar{r}_2$	00000000	00000001		

$$h = 0$$
$$\mathrm{p} = \epsilon_b^h \times (1 - \epsilon_b)^{2w-h}$$

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Algorithm – step 2

$$d = \qquad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$d^* = 00000001$

Lr_{i+1}				
$\bar{r}_1$	00000000	00000001		
$\bar{r}_2$	00000000	00000001		

$$h = 0$$
$$\mathrm{p} = \epsilon_b^h \times (1 - \epsilon_b)^{2w-h}$$
$$\mathrm{p} = 0.15^0 \times 0.85^2$$
$$\mathrm{p} = 0.7225$$

Algorithm – step 2

$$d = \qquad\qquad\qquad\quad \texttt{11010011 01000001 01000010 01001111}$$
$$\widetilde{d_1} = \texttt{11001110 10010011 01010100 11001111 00101101}$$
$$\widetilde{d_2} = \texttt{10000001 11000111 01100010 01111001 11001010}$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$$d^* = \texttt{00000001}$$

Lr_{i+1}				
$\bar{r}_1$	00000000 0.7225	00000001		
$\bar{r}_2$	00000000	00000001		

$$h = 0$$
$$\mathrm{p} = \epsilon_b^h \times (1 - \epsilon_b)^{2w-h}$$
$$\mathrm{p} = 0.15^0 \times 0.85^2$$
$$\mathrm{p} = 0.7225$$

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Algorithm – step 2

$$d = \quad\quad\quad\quad\quad\quad 11010011\ 01000001\ 01000010\ 01001111$$

$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$

$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$d^* = 00000001$

Lr_{i+1}				
$\bar{r}_1$	00000000 0.7225	00000001		
$\bar{r}_2$	00000000	00000001		

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Algorithm – step 2

$$d = \quad\quad\quad\quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$d^* = 00000001$

Lr_{i+1}				
$\overline{\mathrm{r}_1}$	00000000 0.7225	00000001 0.0225		
$\overline{\mathrm{r}_2}$	00000000	00000001		

Algorithm – step 2

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$

$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$

$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$d^* = 00000001$

Lr_{i+1}				
$\overline{r_1}$	00000000 0.7225	00000001 0.0225		
$\overline{r_2}$	00000000 0.0225	00000001		

MUNI

Algorithm – step 2

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$

$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$

$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$d^* = 00000001$

Lr_{i+1}				
$\overline{\mathrm{r}_1}$	00000000 0.7225	00000001 0.0225		
$\overline{\mathrm{r}_2}$	00000000 0.0225	00000001 0.7225		

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Algorithm – step 2

$$d = \qquad\qquad\qquad 11010011\ 01000001\ 01000010\ 01001111$$

$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$

$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$$d^* = 00000001$$

Lr_{i+1}				
$\bar{\mathrm{r}}_1$	00000000 0.7225	00000001 0.0225		
$\bar{\mathrm{r}}_2$	00000000 0.0225	00000001 0.7225		

MUNI

Algorithm – step 2

$$d = \quad\quad\quad\quad 11010011\ 01000001\ 01000010\ 01001111$$

$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$

$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$$d^* = 00000001$$

Lr_{i+1}				
$\bar{\mathrm{r}}_1$	00000000 0.7225	00000001 0.0225		
$\bar{\mathrm{r}}_2$	00000001 0.7225	00000000 0.0225		

Algorithm – step 2

$$d = \quad 11010011\ 01000001\ 01000010\ 01001111$$
$$\widetilde{d_1} = 11001110\ 10010011\ 01010100\ 11001111\ 00101101$$
$$\widetilde{d_2} = 10000001\ 11000111\ 01100010\ 01111001\ 11001010$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

$d^* = 00000001$

	Lr_{i+1}			
$\bar{\mathrm{r}}_1$	00000000 0.7225	00000001 0.0225		
$\bar{\mathrm{r}}_2$	00000001 0.7225	00000000 0.0225		

L

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Algorithm – post step 2

$$d = \quad \texttt{11010011 01000001 01000010 01001111}$$

$$\widetilde{d_1} = \texttt{11001110 10010011 01010100 11001111 00101101}$$

$$\widetilde{d_2} = \texttt{10000001 11000111 01100010 01111001 11001010}$$

iteration $i = 1$, window $\mathrm{t} = 3$, preserved canidates $\mathrm{L} = 2$

Lr_{i+1}		
$\tilde{\mathrm{r}}_1$	00000000	00000001
$\tilde{\mathrm{r}}_2$	00000000	00000001

$$d^* = \texttt{00000001}$$